AI Inference Distribution Network

Host: SONG Jian (China Mobile)
Co-chair: Arashmid Akhavain (Huawei Canada)

Note Well

https://www.ietf.org/about/note-well/

This is a reminder of IETF policies in effect on various topics such as patents or code of conduct. It is only meant to point you in the right direction. Exceptions may apply. The IETF's patent policy and the definition of an IETF "contribution" and "participation" are set forth in BCP 79; please read it carefully.

As a reminder:

- By participating in the IETF, you agree to follow IETF processes and policies.
- If you are aware that any IETF contribution is covered by patents or patent applications that are owned or controlled by you or your sponsor, you must disclose that fact, or not participate in the discussion.
- As a participant in or attendee to any IETF activity you acknowledge that written, audio, video, and photographic records of meetings may be made public.
- Personal information that you provide to IETF will be handled in accordance with the IETF Privacy Statement.
- As a participant or attendee, you agree to work respectfully with other participants; please contact the ombudsteam (https://www.ietf.org/contact/ombudsteam/) if you have questions or concerns about this.

Definitive information is in the documents listed below and other IETF BCPs. For advice, please talk to WG chairs or ADs:

- BCP 9 (Internet Standards Process)
- BCP 25 (Working Group processes)
- BCP 25 (Anti-Harassment Procedures)
- BCP 54 (Code of Conduct)
- BCP 78 (Copyright)
- BCP 79 (Patents, Participation)
- https://www.ietf.org/privacy-policy/ (Privacy Policy)

I E T F

Note to all

- Open to all
- Meeting will be recorded, and all attendees have agreed
- Not under NDA of any form
- Please list your name and affiliation in the chat feature of webex during meeting
- Meeting link: https://ietf.webex.com/meet/sidemeetings2
- Github: https://github.com/jack-songjian-ietf/ietf125-sidemeeting-AI-IDN

I E T F

Motivation

- Two Steps of AI: training and inference
  - Training: building models
  - Inference: applying models to emerges value in real word ← We focus

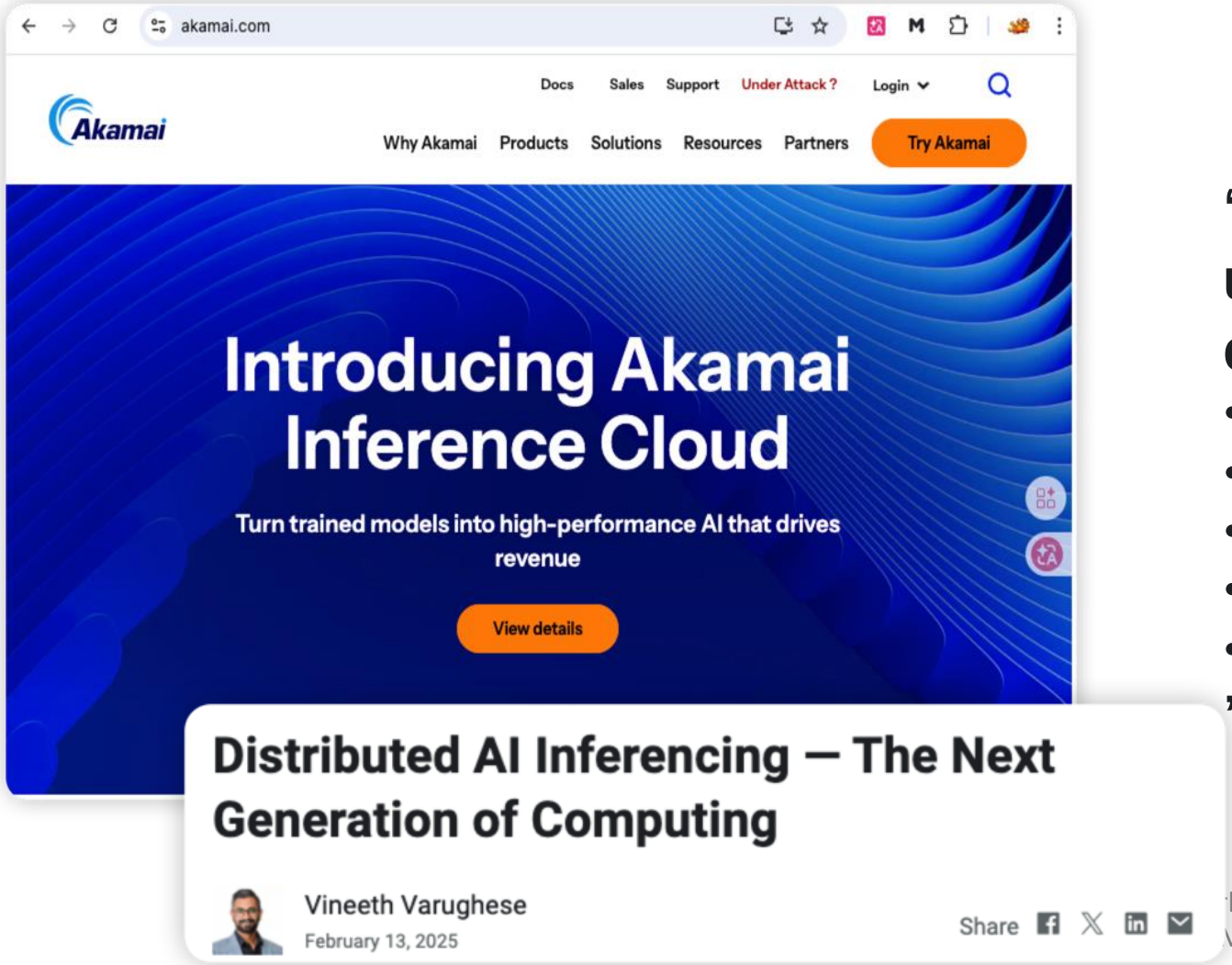
- "AI Inference Distribution Network" is dedicated to discuss:
  - AI inference scenarios (*e.g. inference with privacy protection in medical industry*)
  - distributed AI inference network architectures (*e.g. DIN, DA-ITN, AI services Arch*)
  - models distribution mechanisms
  - inference jobs scheduling mechanisms (*e.g. semantic inference routing protocol*)

  and related technologies (*e.g. edge-cloud collaborative, p/d disaggregation*) that enable efficient and scalable deployment.

Agenda

Start	End	Time (mins)	Toptic	Speaker	Company
09:30	09:40	10	Host	SONG Jian *Arashmid Akhavain*	China Mobile Huawei Canada
09:40	09:55	15	*The Demand and Considerations for Distributed Inference in the Healthcare Industry*	*HUANG Weihong*	Xiangya Hospital
09:55	10:10	15	*Semantic Inference Routing Protocol (SIRP)*	*Huamin Chen*	Redhat
10:10	10:25	15	*Considerations of network/system for AI services*	*Yong-Geun Hong*	Daejeon University
10:25	10:40	15	*AI Network: An overlay network for consumers and providers in AI Ecosystem*	*Arashmid Akhavain*	Huawei Canada
10:40	10:50	10	*Scalable, Secure, Cost-Effective: An End-to-End Edge-Cloud Distributed Solution for Enterprise AI Inference*	*Wenjie Chu*	Huawei
10:50	11:00	10	Open Discussion & Summary	All	

Akamai: Inference is the future of AI



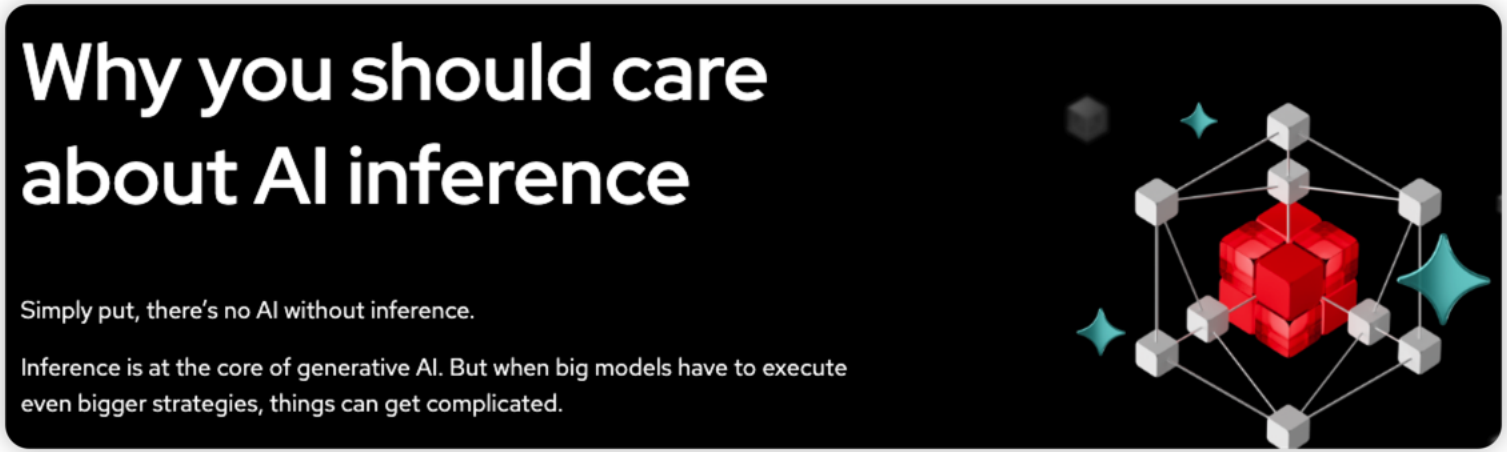
“

unique considerations when delivering AI inference:

- Latency and responsiveness
- Edge device resource constraints
- Security and data privacy
- Scalability and distributed architecture
- Bandwidth and cost efficiency

”

Redhat: Speed, scale, and savings with distributed inference



Challenges of distributed inference:

- Latency and bandwidth
- Resource allocation inefficiencies
- Fault recovery
- Debugging and troubleshooting complexity
- Synchronization overhead
- Model management and deployment
- Cost management
- Security

Advantages of distributed inference:

- Consistent and predictable performance
- Cost management
- Enhanced observability
- Privacy regulations
- Data locality

Open Discussion – IETF Role in AI Inference Distribution

Background

- IETF123: HotRFC presentation “DIN”, Side meeting “DIN & IOA”
- IETF124: rtgwg presentation “DIN usecases-requirements”
- IETF125: Side meeting “AI Inference Distribution Network”

Discussion points

- Use cases & requirements:
  - privacy, security, scalable, concurrency, real time, …
- Architectural Considerations:
  - How to decompose inference tasks across difference nodes?
  - Need for a reference architecture integrating with deep edge computing?
- Technical Challenges
  - Dynamic model selection based on inference job & semantic & network conditions
  - Privacy, security, trust in distributed & collaborative inference
  - Billions level concurrency, e2e low latency for real time inference
- Proposed Next Steps
  - IETF 126: Gather more use cases & requirements & inference technologies & implementation (call for contributions); ;
  - enough interests for a potential BoF?
  - Your input is valuable – let‘s discuss!

Thank you for your attention!

Song Jian

songjianyjy@chinamobile.com